

Week 2 Module

Question 01

Question - 1 :-

- User gives number of Trans. ID
- Store ID in array.
- Take input of suspicious transacti ID
- Traverse the array.
- If found print **Fraud Transaction found**
- else - **Transaction not found.**

```
linear_search.java
1  import java.util.Scanner;
2
3  public class linear_search {
4      public static void main(String[] args) {
5          Scanner sc = new Scanner(System.in);
6
7          int n = sc.nextInt();
8          int[] arr = new int[n];
9
10         for (int i = 0; i < n; i++) {
11             arr[i] = sc.nextInt();
12         }
13
14         int target = sc.nextInt();
15         boolean found = false;
16
17         for (int i = 0; i < n; i++) {
18             if (arr[i] == target) {
19                 found = true;
20                 break;
21             }
22         }
23
24         if (found)
25             System.out.println("Fraud Transaction Found");
26         else
27             System.out.println("Transaction Not Found");
28     }
29 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
Windows PowerShell
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PS C:\Users\Neeraj\Desktop\college\code> cd "c:\Users\N
_search }
5
1201 1305 1450 1408 1502
1450
Fraud Transaction Found
PS C:\Users\Neeraj\Desktop\college\code> |
```

Question 02

Question - 2 :-

- Take input from user
- Store them in array.
- Input of Target.
- $low = 0$ & $high = N-1$
- Find middle element Compare with target
- If found print - **Log found.**

```
Binary_search.java
1  import java.util.Scanner;
2
3  public class Binary_search {
4      public static void main(String[] args) {
5
6          Scanner sc = new Scanner(System.in);
7
8          int n = sc.nextInt();
9          int[] arr = new int[n];
10
11         for (int i = 0; i < n; i++) {
12             arr[i] = sc.nextInt();
13         }
14
15         int target = sc.nextInt();
16
17         int low = 0;
18         int high = n - 1;
19         boolean found = false;
20
21         while (low <= high) {
22
23             int mid = (low + high) / 2;
24
25             if (arr[mid] == target) {
26                 found = true;
27                 break;
28             } else if (arr[mid] < target) {
29                 low = mid + 1;
30             } else {
31                 high = mid - 1;
32             }
33         }
34
35         if (found)
36             System.out.println("Log Found");
37         else
38             System.out.println("Log Not Found");
39     }
40 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS C:\Users\Neeraj\Desktop\college\code> cd "c:\Users\Neeraj\
5
1001 1005 1010 1015 1020
1015
Log Found
PS C:\Users\Neeraj\Desktop\college\code> |
```

Question 03

Question - 3

- take input of Salaries
- Store them in array
- Compare with adjacent element.
- Swap if they are in wrong order
- Repeat until no more swap needed
- Print final array.

```
bubble_sort.java
1  import java.util.Scanner;
2
3  public class bubble_sort {
4
5      public static void main(String[] args) {
6
7          Scanner sc = new Scanner(System.in);
8
9          int n = sc.nextInt();
10         int[] arr = new int[n];
11
12         for (int i = 0; i < n; i++)
13             arr[i] = sc.nextInt();
14
15         for (int i = 0; i < n - 1; i++) {
16
17             for (int j = 0; j < n - i - 1; j++) {
18
19                 if (arr[j] > arr[j + 1]) {
20
21                     int temp = arr[j];
22                     arr[j] = arr[j + 1];
23                     arr[j + 1] = temp;
24
25                 }
26             }
27         }
28
29         for (int i = 0; i < n; i++)
30             System.out.print(arr[i] + " ");
31     }
32 }
33
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
ort }
5
45000 32000 55000 28000 60000
28000 32000 45000 55000 60000 PS C:\Users\Neeraj\Desktop\college\code>
```

Question 04

Question - 4

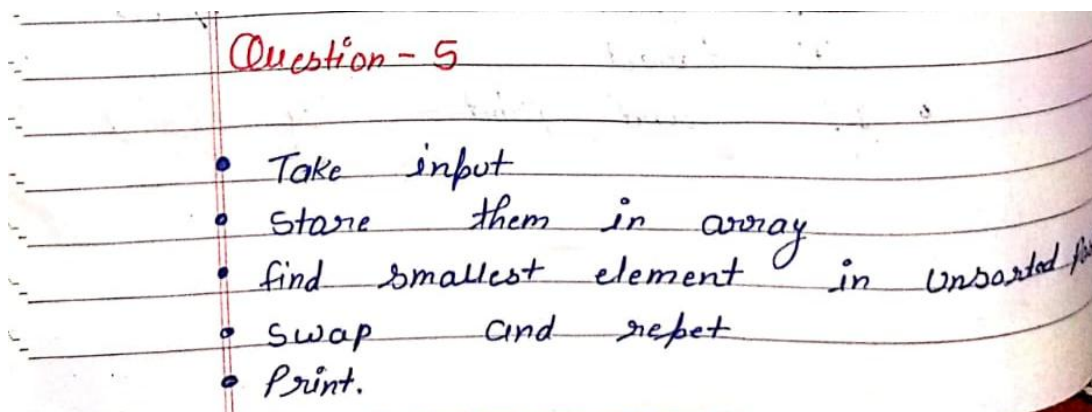
- Take input
- Store them in array.
- let first element is sorted
- pick one element and place in correct position.
- Repeat
- print final array.

```
1 insertion_sort.java
2 import java.util.Scanner;
3 public class insertion_sort {
4
5     public static void main(String[] args) {
6
7         Scanner sc = new Scanner(System.in);
8
9         int n = sc.nextInt();
10
11         int[] arr = new int[n];
12
13         for (int i = 0; i < n; i++)
14             arr[i] = sc.nextInt();
15
16         for (int i = 1; i < n; i++) {
17
18             int key = arr[i];
19             int j = i - 1;
20
21             while (j >= 0 && arr[j] > key) {
22
23                 arr[j + 1] = arr[j];
24                 j--;
25             }
26
27             arr[j + 1] = key;
28         }
29
30         for (int i = 0; i < n; i++)
31             System.out.print(arr[i] + " ");
32     }
33 }
34
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
45000 32000 55000 28000 60000
28000 32000 45000 55000 60000 PS C:\Users\Weeraj\Desktop\college\code>
.java } ; if ($?) { java insertion_sort }
5
250 120 320 180 100
100 120 180 250 320 PS C:\Users\Weeraj\Desktop\college\code>
```

Question 05



```
selection_sort.java
1  import java.util.Scanner;
2
3  public class selection_sort {
4
5      public static void main(String[] args) {
6
7          Scanner sc = new Scanner(System.in);
8
9          int n = sc.nextInt();
10
11         int[] arr = new int[n];
12
13         for (int i = 0; i < n; i++)
14             arr[i] = sc.nextInt();
15
16         for (int i = 0; i < n - 1; i++) {
17
18             int min = i;
19
20             for (int j = i + 1; j < n; j++) {
21
22                 if (arr[j] < arr[min]) {
23                     min = j;
24                 }
25             }
26
27             int temp = arr[i];
28             arr[i] = arr[min];
29             arr[min] = temp;
30         }
31
32         for (int i = 0; i < n; i++)
33             System.out.print(arr[i] + " ");
34     }
35 }
36
```

```
if ($?) { java selection_sort }
5
9 2 8 1 5
1 2 5 8 9 PS C:\Users\Neeraj\Desktop\college\code>
```